

MATH 375: Probability Theory

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Lecture 4

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1 Lecture 4: Conditional Probability and Bayes' Formula

1.1 The birthday paradox

Suppose n people have birthdays that are independent and equally likely to fall on any of 365 days. We ignore February 29.

It is easiest to count the event that *everyone has a different birthday*. The first person can have any birthday. The second must choose one of 364 remaining days, the third one of 363, and so on:

$$P(\text{all birthdays different}) = \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{365 - n + 1}{365}, \quad n \leq 365.$$

Thus, by the complement rule,

$$P(\text{at least two birthdays match}) = 1 - \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n}.$$

With 23 people, the probability of a shared birthday is greater than $\frac{1}{2}$.

1.2 Mutually exclusive events

Suppose A and B cannot occur together, with $P(A) = 0.3$ and $P(B) = 0.5$. Since $A \cap B = \emptyset$,

$$P(A \cup B) = P(A) + P(B) = 0.8.$$

Also,

$$P(A \cap B^c) = P(A) = 0.3, \quad P(A \cap B) = 0.$$

The event A already guarantees that B does not occur.

1.3 Example: which die is larger?

Roll two distinguishable fair dice. There are 36 equally likely ordered outcomes.

Six outcomes are ties, so

$$P(\text{tie}) = \frac{6}{36} = \frac{1}{6}.$$

The remaining outcomes split evenly between “first die is larger” and “second die is larger.” Therefore,

$$P(\text{second die is larger}) = \frac{1 - P(\text{tie})}{2} = \frac{5}{12}.$$

1.4 Drawing two cards

Draw two cards from a standard 52-card deck, without replacement. Because the order does not matter for these questions, the sample space has $\binom{52}{2}$ equally likely pairs.

Both cards are aces. Choose both cards from the four aces:

$$P(\text{both aces}) = \frac{\binom{4}{2}}{\binom{52}{2}}.$$

Both cards have the same rank. Choose one of the 13 ranks, then choose two of its four suits:

$$P(\text{same rank}) = \frac{13\binom{4}{2}}{\binom{52}{2}}.$$

1.5 Example: red socks

A drawer has n socks, three of which are red. If two socks are selected uniformly without replacement and the probability that both are red is $\frac{1}{2}$, find n .

There are $\binom{3}{2}$ favorable pairs out of $\binom{n}{2}$ possible pairs:

$$\frac{\binom{3}{2}}{\binom{n}{2}} = \frac{1}{2}.$$

Since $\binom{3}{2} = 3$ and $\binom{n}{2} = \frac{n(n-1)}{2}$,

$$\frac{6}{n(n-1)} = \frac{1}{2} \implies n(n-1) = 12.$$

The positive solution is

$$n = 4.$$

1.6 Conditional probability

The phrase “**given** A ” tells us to restrict attention to outcomes in A . If $P(A) > 0$, the probability of B given A is

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}.$$

Equivalently, the multiplication rule is

$$P(A \cap B) = P(A)P(B \mid A).$$

Example: two coin flips. Flip a fair coin twice. Let

$$A = \{\text{first flip is heads}\}, \quad B = \{\text{second flip is heads}\}.$$

The four equally likely outcomes are

$$HH, HT, TH, TT.$$

Given A , only HH and HT remain. Of these two outcomes, HH has both flips heads. Hence

$$P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{1/4}{1/2} = \boxed{\frac{1}{2}}.$$

Now suppose instead we are given that *at least one flip is heads*. Only HH , HT , and TH remain. Exactly one of these has both flips heads, so

$$\boxed{P(\text{both heads} | \text{at least one head}) = \frac{1}{3}}.$$

1.7 Bayes' formula

The multiplication rule can be written in either order:

$$P(A \cap B) = P(A)P(B | A) = P(B)P(A | B).$$

Solving for $P(A | B)$ gives **Bayes' formula**:

$$\boxed{P(A | B) = \frac{P(A)P(B | A)}{P(B)}, \quad P(B) > 0.}$$

1.8 Example: knowing versus guessing

A student either knows an answer or guesses. Let

$$K = \{\text{student knows the answer}\}, \quad C = \{\text{student answers correctly}\}.$$

Suppose $P(K) = p$. If the student knows, the answer is correct. If the student guesses among m choices, the answer is correct with probability $1/m$. Thus,

$$P(C | K) = 1, \quad P(C | K^c) = \frac{1}{m}.$$

A correct answer can come from either knowing or guessing:

$$\begin{aligned} P(C) &= P(K)P(C | K) + P(K^c)P(C | K^c) \\ &= p + \frac{1-p}{m}. \end{aligned}$$

Bayes' formula now gives the probability that the student knew the answer, *given that the answer was correct*:

$$\boxed{P(K | C) = \frac{p}{p + (1-p)/m} = \frac{mp}{mp + 1 - p}}.$$